Haoyang Tang

tanghy@idsse.ac.cn | Sanya, Hainan, China | github.com/wojiaonioo | wojiaonioo.github.io

M.S. student in physical oceanography | Differentiable ocean models, GPU acceleration, variational data assimilation and hybrid modeling

Education

Institute of Deep-sea Science and Engineering, Chinese Academy of Sciences **Physical Oceanography, M.S.** 2025.09 - present

Lanzhou University **Atmospheric Sciences, B.S.** 2019.09 - 2023.06

Project

DEFgcm: a differentiable GPU ocean model (sole developer) github.com/wojiaonioo/DEFgcm | 2026.03 - present

The adjoint of MITgcm is generated from its Fortran code and runs on CPUs, and it cannot train large networks inside the time steps end to end. DEFgcm implements the ocean discretizations of MITgcm in PyTorch so that one program computes integrations and their gradients on GPUs, for 4D-Var data assimilation and online training of networks.

Model development and validation

- Hydrostatic and nonhydrostatic Boussinesq equations on latitude–longitude and cubed-sphere grids, with KPP vertical mixing, a pressure-dependent equation of state, an implicit free surface and open boundaries.
- In 29-day South China Sea runs on 960×912×90 and 160×152×90 grids, temperature, salinity, velocity and SSH differ from MITgcm about as much as two MITgcm runs that differ only in the order of their global sums.

GPU acceleration

- On a 960×960×90 grid, four A800 GPUs take 77 ms per step, 1.6 times as fast as MITgcm on 768 Xeon Gold 6132 cores; from N=480 on they are faster than MITgcm on every N×N×90 grid, and a nonhydrostatic step takes a sixth of the MITgcm time on 400 cores.
- Fused the tensor operations of the program into GPU kernels with torch.compile and wrote CUDA kernels for the vertical solves, the equation of state, mixing and advection, with backward kernels for some of them.
- Implemented horizontal domain decomposition and its reverse, in which the halo exchanges add the derivatives back to the owning process, so grids that need several GPUs can be differentiated; at N=960 each doubling of the GPUs is 1.8 times faster, and four GPUs integrate a 2400×2400×90 grid.
- Stored a checkpoint per segment and recomputed the segments in the reverse pass, with the states beyond the GPU budget in host memory; a gradient costs 2.9 to 5.4 times the forward integration.

Variational data assimilation and hybrid modeling

- 4D-Var with the model gradient in South China Sea twin experiments: over 79 three-day windows, the forecast error three days after the window falls by 40% on average, 20% to 57% in individual windows.
- The same reverse integration trains a network that produces the initial state, with the network parameters in place of the initial state as the controls and no target analyses: on 13 test windows it recovers 81% of the 4D-Var improvement from one gradient evaluation, against 44% for a gradient step at the same cost.
- Differentiated a network that adds a momentum tendency inside every time step: the weight gradient agrees with central differences to a relative 10^{-6}, at 8% extra time per gradient.

Skills

Programming: PyTorch (torch.compile, custom operators, autograd), CUDA C++, Python, Fortran, NCCL / MPI
Scientific computing: ocean numerical models, automatic differentiation and adjoints, variational data assimilation, hybrid modeling
Tools: Git, SLURM, Nsight, Claude Code, Codex